

Radiation from the Compact Fiber: A Unified Architecture for Polarization, Interference, Correlation, Topology, and Radiation–Matter Transfer

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Abstract

This paper develops the radiation sector of the compact-fiber framework and argues that a broad benchmark class of radiation phenomena can be recovered from one radiation architecture together with the already-established Paper III representational/readout layer. The benchmark regime treated here includes polarization, single-photon interference, Bell polarization correlation, Hong–Ou–Mandel bunching, orbital angular structure, the photoelectric event architecture, and the associated scaling bridge to conventional units.

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1 Introduction

Standard radiation and quantum-optical phenomena are usually presented through a collection of separate laws, calculational rules, and formal devices. Polarization, interference, Bell correlation, Hong–Ou–Mandel bunching, orbital angular structure, and the photoelectric effect are commonly treated through partially overlapping but still distinct theoretical entry points. Even when those results are gathered under a common Hilbert-space formalism, the framework is typically introduced at the representational level, with the underlying structural content left primitive.

We claim that a broad benchmark class of radiation phenomena can be recovered from one compact-fiber radiation architecture together with the already-established representational and readout layer developed in Paper III. The claim is not that all radiation physics is completed here, nor that every optical or quantum-electrodynamic regime is addressed. The claim is narrower and sharper: within the benchmark regime of passive linear radiation phenomena, together with the core radiation–matter transfer benchmark, one common structural backbone is sufficient to recover the main benchmark laws without introducing disconnected phenomenon-specific rules.

We also claim that this framework compares favorably with current legacy treatments of the same benchmark regime on three related axes: parsimony, structural continuity, and derivational grounding. Standard benchmark-by-benchmark presentations typically begin by assuming analyzer laws, superposition rules, singlet states, beamsplitter phase conventions, angular mode families, threshold formulas, or the primitivity of the Planck constant. The present framework instead derives or grounds those benchmark ingredients within one radiation architecture and one shared representational/readout layer.

The framework also makes explicit predictive and falsifiable commitments. Some results are recoveries of standard benchmark laws, but they are recoveries with reduced primitive assumptions. Others are stronger structural commitments: the OAM parity selection rule, the readout-class coherence criterion, the atomic photoelectric threshold bridge, and the existence of a structured scaling bridge between recurrence-based natural content and conventional units. The framework is therefore not merely an alternative vocabulary for familiar results; it commits itself to specific structural and quantitative consequences that can fail.

We therefore present this paper as a foundational reference for the compact-fiber radiation benchmark regime: broad enough to unify the main benchmark classes, but bounded enough to remain structurally disciplined.

2 Scope, regime, and claim strength

This paper is about the compact-fiber radiation benchmark regime. It does not attempt to present a complete radiation theory, a full reformulation of quantum electrodynamics, or a treatment of every optical or many-photon phenomenon. Its scope is narrower: to establish one compact-fiber radiation architecture, together with the already-established Paper III representational/readout layer, as sufficient for a broad benchmark class of passive linear radiation phenomena plus the core radiation–matter transfer benchmark.

In scope

The paper directly treats the following benchmark classes:

- single-photon polarization readout and Malus’s law,
- single-photon interference, including Mach–Zehnder interferometry,

- Bell polarization correlation,
- Hong–Ou–Mandel bunching,
- orbital angular structure at the level developed here,
- photoelectric event architecture,
- the radiation network calculus that unifies the passive linear benchmarks,
- Planck scaling as the natural-to-conventional unit bridge used across the radiation program.

These are treated as one connected benchmark suite, not as isolated case studies.

Out of scope

The following are intentionally excluded from the present paper:

- a purely fiber-level resolution of every integer ℓ without the representational layer,
- solid-state work-function derivation from first principles,
- nonlinear optics,
- gain media and lasing,
- full Fock-space or variable-photon-number treatment,
- antibunching,
- delayed-choice quantum eraser,
- full spin-orbit coupling,
- mass-anchor internalization in the Planck scaling layer.

Some of these are outside the regime of the present paper. Others are plausible follow-on problems that would weaken this paper if only partially developed here.

Regime statement

The passive linear part of the paper assumes mode transport and junction composition without nonlinear coupling, gain, or variable-photon-number dynamics. The radiation–matter transfer section treats the photoelectric effect at the level of event architecture, threshold structure, and the atomic threshold bridge, not as a complete many-body solid-state theory.

Claim-strength convention

To keep the paper disciplined, benchmark results are classified using the following labels:

Label	Meaning
Closed	The benchmark law or architecture is derived strongly enough for the paper’s main claim.
Closed at benchmark level	The benchmark is structurally recovered for the paper’s intended scope, but not elevated to a broader universal theorem.
Closed with representational completion	The fiber fixes the quantization or coarse structure, while the representational layer resolves the full continuous or integer detail.
Architecture-level	The event or structural logic is established, but some quantitative target values remain empirical.
Excluded	Intentionally outside the regime of the present paper.

These labels are not rhetorical. They are part of the paper’s logic. They distinguish what is claimed strongly, what is claimed only within the benchmark regime, and what is deliberately not claimed here.

Present status of the main benchmarks

At the level used in this paper, the benchmark statuses are:

- polarization / Malus — **Closed**,
- single-photon interference — **Closed**,
- Bell — **Closed**,
- HOM — **Closed**,
- OAM — **Closed with representational completion**,
- photoelectric — **Closed at benchmark level**, strongest for atomic targets,
- Planck scaling — **Closed at benchmark level**, with structural form fixed and two-anchor scaling.

This means the paper’s central claim does not rest on unresolved benchmark physics. The remaining limits concern regime boundaries, scope control, and deeper follow-on questions, not the viability of the main benchmark suite itself.

Paper-level standard

A section belongs in the main paper only if it supports the central claim at its present strength. If a result is weaker than that, it should either be moved to an appendix, stated explicitly as out of scope, or deferred to later work. That rule is what keeps the paper broad without becoming diffuse.

3 Framework

Paper I established the compact fiber as the admissibility structure of the broader framework. Paper III established the representational and readout layer through which effective quantum-mechanical behavior is recovered. The relevant Paper III result is the compact-fiber representational layer developed in [7]. The present paper develops the radiation sector that results when those foundations are applied systematically to the radiation benchmark regime. Radiation is therefore not introduced here as an unrelated special case, nor is it assumed to be self-evident from Paper III alone. What the present paper contributes is the explicit organization of the radiation branch as a unified benchmark architecture.

The framework used throughout the paper has four components. It begins with a radiation-side structural hierarchy. It then reuses the Paper III representational and readout layer. It fixes the natural-unit scale through a minimal empirical input ledger. Finally, it relates that scale to conventional energy and frequency units through the Planck bridge discussed later in the paper. The present section states those ingredients in the form needed for the benchmark sections that follow.

3.1 Radiation as a sector of the framework

The present paper treats radiation as an explicit sector of the compact-fiber program. The benchmark sections should therefore not be read as a sequence of unrelated applications. They are intended as manifestations of one radiation-side development of the framework. The paper’s central architectural claim is that the benchmark regime can be recovered from one radiation hierarchy together with one shared representational/readout layer, rather than from a collection of phenomenon-specific starting assumptions.

The radiation-side ontology begins from coherent return. A radiation mode is not treated as a primitive field oscillation introduced without further structural account. It is treated as a recurrence-capable structure whose observable distinctions arise through successive levels of admissible organization. The hierarchy summarized in [table 1](#) proceeds from differentiated coherent return through aspect and sign structure to covering monodromy. These levels are not separate physical entities. They are successive enrichments of one radiation architecture.

Level	Structural content	First benchmark role
Level 1	differentiated coherent return	transport, recurrence, branch propagation, interference substrate
Level 2	aspect structure	transverse directional organization, polarization-type decomposition
Level 3	sign structure	analyzer branching, opposition, sign-bearing readout structure
Level 4	covering-monodromy structure	topological parity and angular residue structure

Table 1: Radiation-side structural hierarchy used in the benchmark regime.

The hierarchy in [table 1](#) should be read as a sequence of enrichments of one radiation architecture rather than as a list of separate physical entities. Its role in the paper is cumulative: Level 1 supplies

the common recurrence-bearing substrate, Levels 2 and 3 add the transverse and sign-bearing structure needed for polarization and analyzer-based benchmarks, and Level 4 adds the covering-monodromy structure used in the OAM result. The benchmark sequence is therefore ordered not only expositively but also by increasing structural depth.

The benchmark sequence can be read in terms of the minimum structural depth required in each case. Polarization depends on Level 2/3 structure. Single-photon interference depends primarily on Level 1 together with branch transport. Bell uses Level 2/3 structure together with non-factorable pair structure in the representational layer. HOM uses the same radiation structure together with tensor-product and exchange-symmetry machinery. OAM uses Level 4 together with representational completion. Photoelectric transfer depends most directly on Level 1 together with target-side binding structure. In this sense, the order of the benchmarks is not merely expository. It also tracks the increasing structural demands placed on the radiation architecture.

3.2 Representational and readout layer

The compact fiber does not by itself provide the full calculational machinery used for projection, probability assignment, or multi-particle benchmark calculations. Those roles are supplied by the Paper III representational layer [7], and the present paper uses that layer throughout as a stated prerequisite rather than re-proving it internally.

For the purposes of the present radiation benchmark paper, the load-bearing Paper III results are these. First, compact-fiber states admit an effective Hilbert-space realization in which basis-dependent projections can be defined. Second, the Born-rule layer provides the shared readout law used for all benchmark probabilities in the present paper. Third, the same representational layer supplies the tensor-product structure required for two-arm and two-photon benchmarks. Fourth, it supplies the exchange-symmetry machinery used in the HOM analysis. The present paper therefore does not appeal to Paper III in a vague or purely programmatic sense; it relies on a definite set of representational and readout results already established there.

The division of labor should be read accordingly. Paper I establishes the admissibility structure. Paper III establishes the effective representational and readout machinery. The present paper develops the radiation sector that results when those ingredients are applied systematically to the benchmark regime treated here. In particular, the benchmark sections do not introduce separate readout laws for polarization, interference, Bell, HOM, or photoelectric transfer. They reuse the same representational/readout layer across the suite.

This division of labor is part of the paper’s parsimony claim. The point is not to avoid standard mathematics. The point is to show that one radiation architecture together with one already-established representational/readout layer is sufficient to support the benchmark suite, rather than requiring a separate starting formalism for each phenomenon.

3.3 Minimal empirical input ledger

The framework uses two dimensionally independent empirical anchors to set the natural-unit scale: one spectroscopic anchor for the length side and one mass/gravitational anchor for the mass side. The speed of light is exact by convention. Structural ratios, benchmark laws, and selection rules are then fixed by the compact-fiber architecture together with the Paper III representational/readout layer.

This statement is part of the paper’s parsimony claim. The framework does not treat every benchmark constant as primitive. It distinguishes what is structural, what is calibrated, and what

is exact by convention. That distinction becomes especially important in the scaling discussion later in the paper.

3.4 Forward relation to Planck scaling

The Planck relation $E = h\nu$ is treated in this framework as structurally required in form, not as an independent postulate imported from outside. The constant h is interpreted as the unit bridge between recurrence-based natural units and conventional measurement units. Its canonical scaling form is recorded in [appendix A](#).

For present purposes, the important point is that the benchmark sections do not simply assume h as an unexplained primitive. They proceed within a framework in which recurrence content is primary and conventional energy units arise only after the natural-to-conventional scaling bridge is applied.

The rest of the paper proceeds with the framework fixed in this form. Radiation is treated as an explicit sector of the compact-fiber program rather than as an assumed corollary. That sector is organized by a four-level structural hierarchy, while Paper III supplies the shared representational and readout layer. The natural-unit scale is fixed by a minimal empirical input ledger, and the Planck relation is reserved for later scaling analysis rather than assumed at the outset. With that structure in place, the first benchmark can be stated directly: single-photon polarization readout.

4 Polarization benchmark

The first benchmark should be the simplest one that directly exposes the joint use of the radiation hierarchy and the shared readout layer. That benchmark is single-photon polarization readout. It asks whether the compact-fiber radiation architecture, together with the Paper III projection and Born layer, can recover a standard analyzer law without treating polarization as an independent primitive property and without introducing a benchmark-specific measurement rule.

In the present framework, polarization is associated with the Level 2/3 transverse organization of the radiation mode. Level 2 supplies aspect structure, while Level 3 supplies the sign-bearing differentiation that becomes relevant under analyzer decomposition. The present paper does not claim that the Malus calculation is performed directly at the pre-representational level. Rather, it interprets the representational polarization state as the image of a Level 2/3 transverse organization already identified on the radiation side.

In the representational layer, the transverse state at a magnetic base point is expressed in a two-dimensional compatible complex space $V_b \cong \mathbb{C}^1$. A mode with transverse orientation ϕ is therefore written as

$$|\phi\rangle = \cos \phi |e_1\rangle + \sin \phi |e_2\rangle,$$

where $|e_1\rangle$ and $|e_2\rangle$ are the orthonormal basis vectors associated with the canonical transverse orientations.

An analyzer at angle θ is treated as a readout context, not as an unexplained external measurement box. In the representational layer it defines the decomposition

$$|m_\theta\rangle = \cos \theta |e_1\rangle + \sin \theta |e_2\rangle,$$

with orthogonal complement

$$|m_{\theta+\pi/2}\rangle = -\sin \theta |e_1\rangle + \cos \theta |e_2\rangle.$$

The analyzer therefore has a double status. On the radiation side it selects a decomposition of the Level 2/3 transverse organization. In the Paper III layer it is represented as a basis choice for projection and readout.

Once that basis is fixed, the readout law is the same Born-rule layer reused throughout the paper:

$$P(+ | \theta, \phi) = |\langle m_\theta | \phi \rangle|^2.$$

The inner product is

$$\langle m_\theta | \phi \rangle = \cos \theta \cos \phi + \sin \theta \sin \phi = \cos(\theta - \phi),$$

and therefore

$$P(+ | \theta, \phi) = \cos^2(\theta - \phi).$$

This is Malus's law.

The significance of the result is limited but still important. The analyzer law is not inserted here as an independent optical postulate; it is recovered within the shared Paper III projection and Born layer, while being interpreted as the representational image of Level 2/3 transverse structure on the radiation side. The benchmark therefore establishes the division of labor that later sections reuse: radiation-side structure fixes the physical interpretation of the mode, and the Paper III layer supplies the effective projection and readout machinery.

The polarization benchmark therefore establishes the first benchmark-level use of the shared representational/readout layer within the radiation sector. A standard analyzer law is recovered without introducing a polarization-specific extra rule, and the section fixes the division of labor that the later benchmarks make more structurally explicit. Once that first single-arm benchmark is in place, the next task is to make explicit the reusable calculational structure that governs passive linear radiation networks more generally.

5 Radiation network calculus

The polarization benchmark establishes the first direct payoff of the framework, but by itself it does not yet show how the same architecture governs propagation through multi-path or multi-port optical arrangements. The next step is therefore to state the reusable calculational core for the passive linear part of the radiation program. That core is the radiation network calculus.

The calculus does not introduce a new theory layer. It makes explicit the small set of primitives and composition rules already implicit in the passive-network benchmarks. In the regime treated here, passive linear radiation experiments are built from three ingredients: branches, junctions, and readout.

A branch is a directed path together with its transport factor. At the representational level relevant for the passive-network benchmarks, the transport factor is a phase,

$$\text{branch} = (\text{path}, e^{i\phi}).$$

The phase ϕ encodes the transport content accumulated along the path. Its detailed physical interpretation depends on the benchmark. In single-photon interference it appears as relative path phase. In the OAM section it appears as angular phase winding. In more general passive networks it is the phase accumulated between junctions. At a coarser pre-representational level, the same transport may be visible only through residue structure such as sign or parity, but for the network calculus itself the working object is the representational transport factor $e^{i\phi}$.

A junction is a spatial node at which modes split or recombine. The two-port junction used throughout the passive-network benchmarks has transmission amplitude t and reflection amplitude

rJ , where J is the quarter-turn structure already present in the compact fiber. On the fiber side, the general form is

$$U(t, r) = \begin{pmatrix} t & rJ \\ rJ & t \end{pmatrix},$$

and in the complex representation, where $J \mapsto i$, this becomes

$$U(t, r) = \begin{pmatrix} t & ir \\ ir & t \end{pmatrix}.$$

The amplitudes satisfy

$$t^2 + r^2 = 1.$$

With $t, r \in \mathbb{R}$ and $J \mapsto i$, the junction matrix satisfies $U^\dagger U = I$ whenever $t^2 + r^2 = 1$.

This division of roles should be kept clear. The splitting ratio is a property of the junction geometry, whereas the quarter-turn phase is the structural contribution supplied by the compact fiber. In the balanced case, $t = r = 1/\sqrt{2}$, one recovers the standard 50/50 beamsplitter form. More general passive junctions are represented by the same rule with different t and r . The phase J is therefore not introduced here as a free convention chosen only to reproduce textbook optics. In the present radiation sector it is the framework-motivated realization of junction reflection, interpreted through the quarter-turn structure already present in the compact fiber.

The readout rule is the same shared Born layer already used in the polarization benchmark. If multiple coherent branch amplitudes arrive in the same readout class, they combine before squaring:

$$P = |c_{\text{total}}|^2.$$

No separate readout law is introduced for polarization, interference, or HOM. The same readout layer is reused throughout the passive-network regime.

5.1 Readout class

Let D be a detector or readout device, and let Π_D denote the Paper III projection structure associated with its available output channels. Two incident branches b_1 and b_2 are said to belong to the same readout class at D if they are equivalent under Π_D in all detector-resolved degrees of freedom except for phase-carrying structure. Equivalently, b_1 and b_2 belong to the same readout class when D assigns them to the same output channel and cannot distinguish them by any readout-relevant datum other than phase.

In the present benchmark regime, detector-resolved data may include spatial mode, arrival-time window, polarization channel, and any other degree of freedom explicitly resolved by the readout architecture. Phase itself does not split readout class; it is the degree of freedom along which coherent recombination occurs within a class.

Composition enters in two forms. In series composition, a single mode passing through a sequence of paths and junctions is described by matrix multiplication. If U_k are junction matrices and P_k are diagonal phase matrices for the path segments between them, then

$$|\psi_{\text{out}}\rangle = U_n P_n U_{n-1} P_{n-1} \cdots P_1 U_1 |\psi_{\text{in}}\rangle.$$

This is the form instantiated explicitly in the Mach–Zehnder benchmark of the next section. In parallel composition, when multiple radiation modes are simultaneously present, the state space is extended by tensor product. The Paper III layer then supplies the multi-particle machinery, including exchange symmetry where appropriate. That is the route used later in the HOM benchmark.

5.2 Coherence criterion

The coherence criterion can now be stated precisely. Branch amplitudes combine coherently at a detector if and only if they belong to the same readout class there. If their differences are confined to phase-carrying structure, the amplitudes add before squaring. If they differ in detector-resolved data and therefore fall into different readout classes, they do not contribute to the same coherent total.

$$P = |c_\alpha + c_\beta|^2,$$

whereas in the incoherent case

$$P = |c_\alpha|^2 + |c_\beta|^2.$$

The distinction between coherent and incoherent addition is therefore not treated as arbitrary or purely procedural. It is fixed by the readout architecture itself. That criterion becomes load-bearing in the interference benchmark, reappears in HOM through indistinguishability, and later enters the paper’s falsifiability discussion.

The radiation network calculus is the reusable middle layer of the radiation program. It sits between the more primitive radiation hierarchy and the benchmark applications that follow. The next section gives its first full use: single-photon interference as coherent recombination of multiple branches of one radiation process.

6 Single-photon interference benchmark

With the radiation network calculus in place, the first full passive-network benchmark can be stated directly. Single-photon interference is the natural test case because it is the simplest phenomenon that requires branch transport, coherent recombination, and the readout-class distinction between coherent and incoherent addition. In the present framework, interference is not treated as a primitive wave law layered on top of the radiation architecture. It is treated as coherent recombination of multiple branches of one radiation process.

That distinction is central. The primitive object is not two independent events that happen to overlap. It is one radiation process with more than one admissible branch. The interference pattern appears when those branches are recombined under the same readout architecture already introduced in the previous sections.

6.1 One process, multiple branches

A two-path interference setup is modeled as a decomposition of one radiation realization into coherent branches,

$$\Psi \rightarrow \Psi_\alpha + \Psi_\beta.$$

Each branch carries its own transport factor. At the representational level,

$$c_\alpha = |c_\alpha|e^{i\phi_\alpha}, \quad c_\beta = |c_\beta|e^{i\phi_\beta}.$$

The relative branch relation is then the phase difference

$$\Delta = \phi_\alpha - \phi_\beta.$$

The significance of this representation is that the two branches are not treated as primitive independent objects. They are branches of one coherent-return structure. In this sense, the framework treats interference as one-process branching rather than many-process pileup.

6.2 Recombination and Born readout

When the two branches recombine at a detector and belong to the same readout class, their amplitudes add:

$$c_{\text{total}} = c_{\alpha} + c_{\beta}.$$

The readout probability is therefore

$$P = |c_{\alpha} + c_{\beta}|^2.$$

Expanding gives

$$P = |c_{\alpha}|^2 + |c_{\beta}|^2 + 2 \operatorname{Re}(c_{\alpha} \overline{c_{\beta}}),$$

so the interference term appears directly from the general readout law once coherent branch recombination is present.

No benchmark-specific interference law is introduced here. Once one radiation process admits multiple coherent branches and those branches recombine in the same readout class, the cross term follows from the same shared readout layer already used in the polarization benchmark.

6.3 Coherence and decoherence

The interference benchmark is also the first place where the coherence criterion becomes physically load-bearing.

If two branches arrive at the detector in the same readout class and differ only in phase-carrying structure, they contribute to one coherent amplitude sum. If they differ in non-phase data relevant to readout, they do not combine coherently, and the detector registers only incoherent addition. Thus

$$P = |c_{\alpha} + c_{\beta}|^2$$

in the coherent case, whereas

$$P = |c_{\alpha}|^2 + |c_{\beta}|^2$$

in the incoherent case.

This is the framework's structural account of the coherence/decoherence distinction. The question is not merely whether an external environment has recorded which path. The sharper claim is that coherence survives exactly when the branches remain indistinguishable within the readout architecture except for phase-carrying structure.

6.4 Mach–Zehnder as worked case

The standard worked example is the Mach–Zehnder interferometer. It is described by a sequence of one junction, one path-phase matrix, and a second junction:

$$U \cdot P \cdot U.$$

For a balanced 50/50 junction,

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix},$$

and for arm phases ϕ_1 and ϕ_2 ,

$$P = \begin{pmatrix} e^{i\phi_1} & 0 \\ 0 & e^{i\phi_2} \end{pmatrix}.$$

In the convention used here, both junctions are represented by the same symmetric balanced matrix U rather than by U followed by $U^\dagger$, so the output amplitudes below should be read in that sign convention.

A direct computation gives the output amplitudes

$$c_e = \frac{1}{2} (e^{i\phi_1} - e^{i\phi_2}), \quad c_f = \frac{i}{2} (e^{i\phi_1} + e^{i\phi_2}).$$

If $\Delta = \phi_1 - \phi_2$, the output probabilities are

$$P_e = \sin^2(\Delta/2), \quad P_f = \cos^2(\Delta/2).$$

This is the standard Mach–Zehnder result. Its significance here is not only that the familiar trigonometric law is recovered, but that it follows directly from the same branch/junction/readout calculus already defined in [section 5](#).

The interference benchmark supports three claims at once. First, one radiation process may admit multiple coherent branches without introducing a second primitive object. Second, interference is not a special extra law but the direct result of coherent recombination under the shared Born-rule readout layer. Third, the section establishes the coherence criterion that later reappears in HOM and in the paper’s falsifiability discussion.

Single-photon interference is therefore the first full demonstration that the radiation hierarchy and the network calculus together generate a standard quantum-optical phenomenon from one structural scheme. The next benchmark tests whether that same scheme, together with the shared representational layer, also handles non-factorable two-arm correlation rather than only one-process branching.

7 Bell benchmark

The interference benchmark shows that one radiation process can branch and recombine coherently within the shared readout architecture. The next question is whether the same framework also handles non-factorable two-arm correlation. The Bell benchmark is the first test of that stronger requirement. For modern loophole-free photon-polarization Bell benchmarks, see [\[2, 6\]](#).

The benchmark treated here is the standard polarization Bell configuration. Its target is the correlation law

$$E(\alpha, \beta) = -\cos 2(\alpha - \beta),$$

together with the standard CHSH-optimal value

$$|S| = 2\sqrt{2}.$$

The importance of the section is not only that these results are recovered. It is that the framework grounds several ingredients that standard Hilbert-space treatments typically take as given: the pair-state structure, the analyzer contexts, and the reuse of the Born layer.

7.1 State space and analyzer contexts

The radiation-side polarization structure already introduced in the Malus benchmark lifts, via the Paper III layer, to a two-dimensional complex space at each arm. For the two-arm benchmark, the joint state space is therefore

$$V_A \otimes V_B \cong \mathbb{C}^2 \otimes \mathbb{C}^2.$$

Analyzer settings are again readout contexts rather than primitive measurement boxes. At arm A , an analyzer at angle α defines the basis vector

$$|m_\alpha\rangle = \cos \alpha |e_1\rangle + \sin \alpha |e_2\rangle,$$

with orthogonal complement

$$|m_{\alpha+\pi/2}\rangle = -\sin \alpha |e_1\rangle + \cos \alpha |e_2\rangle.$$

The same form holds at arm B for angle β . The two-arm benchmark therefore uses the same analyzer logic already established in the polarization section, now in tensor-product form.

7.2 Opposition condition and unique pair state

The load-bearing radiation-side condition is opposition: for equal analyzer settings, the two readout outcomes must be opposite. In the representational layer, this means that for every analyzer angle θ ,

$$\langle m_\theta, m_\theta | \Psi \rangle = 0.$$

A general joint state in $\mathbb{C}^2 \otimes \mathbb{C}^2$ can be written as

$$|\Psi\rangle = a |e_1\rangle|e_1\rangle + b |e_1\rangle|e_2\rangle + c |e_2\rangle|e_1\rangle + d |e_2\rangle|e_2\rangle.$$

Requiring the equal-context overlap to vanish for all θ forces

$$a = 0, \quad d = 0, \quad b + c = 0.$$

This leaves a one-dimensional subspace. After normalization, the unique opposed pair state is

$$|\Psi_{\text{opp}}\rangle = \frac{1}{\sqrt{2}} (|e_1\rangle|e_2\rangle - |e_2\rangle|e_1\rangle).$$

Standard treatments ordinarily posit the singlet and proceed. The present framework instead forces the opposed pair state from the opposition condition.

7.3 Born probabilities and the correlation law

Once the pair state and analyzer contexts are fixed, the readout law is again the same Born rule reused from earlier sections. For example, the joint probability of equal $+, +$ outcomes is

$$P(+, +) = |\langle m_\alpha | \otimes \langle m_\beta | \Psi_{\text{opp}} \rangle|^2.$$

A direct computation gives

$$P(+, +) = \frac{1}{2} \sin^2(\alpha - \beta).$$

Likewise,

$$P(+, -) = \frac{1}{2} \cos^2(\alpha - \beta),$$

and similarly

$$P(-, -) = \frac{1}{2} \sin^2(\alpha - \beta), \quad P(-, +) = \frac{1}{2} \cos^2(\alpha - \beta).$$

The correlation is therefore

$$E(\alpha, \beta) = P(+, +) + P(-, -) - P(+, -) - P(-, +),$$

which yields

$$E(\alpha, \beta) = \sin^2(\alpha - \beta) - \cos^2(\alpha - \beta) = -\cos 2(\alpha - \beta).$$

7.4 CHSH as a direct corollary

The CHSH-optimal value follows immediately from the recovered cosine law. For the standard settings

$$a = 0, \quad a' = \frac{\pi}{4}, \quad b = \frac{\pi}{8}, \quad b' = \frac{3\pi}{8},$$

the CHSH combination

$$S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$$

gives

$$|S| = 2\sqrt{2}.$$

7.5 Why the cosine law is earned

A deterministic hard-threshold readout route was examined and rejected because it produces the wrong angular dependence: a triangle-wave structure rather than the cosine law. The correct closure comes from the general Born-rule readout architecture acting on the unique opposed pair state, not from a Bell-specific threshold mechanism.

The Bell benchmark contributes four things beyond standard benchmark recovery.

First, the pair state is forced by the opposition condition rather than posited independently.

Second, analyzer contexts are grounded in the same Level 2/3 polarization structure already used in the single-arm benchmark, rather than introduced as a separate Bell-only apparatus formalism.

Third, the Born rule is reused from the shared Paper III layer. No Bell-specific probability law is added.

Fourth, the benchmark demonstrates that one radiation architecture can handle both single-arm polarization readout and two-arm non-factorable correlation without changing its conceptual basis.

The compact-fiber radiation architecture, together with the Paper III readout layer, therefore yields the unique opposed pair state and reproduces the standard Bell polarization correlation law

$$E(\alpha, \beta) = -\cos 2(\alpha - \beta),$$

with the CHSH-optimal value $|S| = 2\sqrt{2}$ following as a direct corollary.

The next section tests a different nonclassical optical structure: not two-arm correlation of a paired state, but indistinguishable two-photon interference at a junction.

8 HOM benchmark

The Bell section established that the framework handles non-factorable two-arm correlation through the same shared readout layer. The next question is whether it also handles indistinguishable two-photon interference. Hong–Ou–Mandel bunching is the natural test, because it requires the junction rule, tensor-product structure, and exchange symmetry to act together within one benchmark. The canonical experimental benchmark is the original Hong–Ou–Mandel interference measurement [3].

HOM is structurally different from the single-photon interference benchmark. Single-photon interference is one-process branching and recombination. HOM requires a joint two-photon state, a junction that mixes two input modes, and the exchange-symmetry structure already established in the Paper III layer. The benchmark therefore tests whether the framework extends from one-photon branch interference to indistinguishable two-photon interference without introducing new primitives.

8.1 Two-photon input and exchange symmetry

The input consists of one photon entering each arm of a two-port junction. At the representational level, the two-photon space is the tensor product of the single-photon mode spaces. Because photons are bosons, the physical two-photon state is symmetric under exchange. If $|a\rangle$ and $|b\rangle$ denote the two input mode states, then the symmetrized input is

$$|\Psi_{\text{in}}\rangle = \frac{1}{\sqrt{2}} (|a\rangle|b\rangle + |b\rangle|a\rangle).$$

This is not an additional HOM-specific rule. The exchange-symmetry sector is already part of the accepted Paper III machinery and is simply being applied here in the radiation benchmark setting.

8.2 The junction rule at a balanced splitter

The relevant spatial device is the balanced two-port junction introduced in the radiation network calculus. In the complex representation, the balanced rule is

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}.$$

Transmission is represented by a real amplitude, while reflection carries the quarter-turn phase $J \mapsto i$. That phase is not inserted by convention to reproduce textbook optics. It is the representational image of the compact-fiber quarter-turn structure already present in the framework.

If the input modes are labeled a and b and the output modes c and d , then the balanced junction acts as

$$|a\rangle \mapsto \frac{1}{\sqrt{2}} (|c\rangle + i|d\rangle), \quad |b\rangle \mapsto \frac{1}{\sqrt{2}} (i|c\rangle + |d\rangle).$$

8.3 Coincidence-sector cancellation

Applying the junction map to the symmetrized two-photon input state gives the output explicitly. Starting from

$$|\Psi_{\text{in}}\rangle = \frac{1}{\sqrt{2}} (|a\rangle|b\rangle + |b\rangle|a\rangle),$$

the junction acts on the two-photon sector by the tensor-product lift of the single-photon map. The two tensor-product terms transform as

$$|a\rangle|b\rangle \mapsto \frac{1}{2} (i|c\rangle|c\rangle + |c\rangle|d\rangle - |d\rangle|c\rangle + i|d\rangle|d\rangle),$$

and

$$|b\rangle|a\rangle \mapsto \frac{1}{2} (i|c\rangle|c\rangle - |c\rangle|d\rangle + |d\rangle|c\rangle + i|d\rangle|d\rangle).$$

Adding the two contributions and dividing by $\sqrt{2}$ yields

$$|\Psi_{\text{out}}\rangle = \frac{i}{\sqrt{2}} (|c\rangle|c\rangle + |d\rangle|d\rangle).$$

The coincidence-sector terms cancel exactly, so no $|c\rangle|d\rangle$ contribution survives. Hence

$$P_{\text{coinc}} = 0.$$

Both photons exit the same port. This is the standard HOM bunching result.

The cancellation is the load-bearing part of the benchmark. It shows that the framework is not simply combining two independent single-photon branch sums. What interferes are exchange-related two-photon amplitudes contributing to the same output sector, and their cancellation is what produces the HOM dip.

8.4 Indistinguishability as readout-class matching

The framework’s interpretation of indistinguishability is one of the main outputs of the section. The photons bunch when they match in readout class at the splitter except for the phase-carrying structure relevant to the joint interference process. In practical terms, this means agreement in the benchmark-relevant data such as frequency, polarization, and arrival-time overlap.

This is the two-photon analogue of the coherence criterion introduced in the interference section. There, coherence survives when two branches remain indistinguishable at the detector except for phase-carrying structure. Here, HOM interference survives when the two photons remain indistinguishable at the splitter in the same structural sense. The distinction between coherence and distinguishability is therefore not phenomenon-specific. It is governed by the readout architecture itself.

8.5 Partial distinguishability

If the two photons are not perfectly matched, the cancellation in the coincidence sector is incomplete. The framework expresses that mismatch through an overlap or visibility factor. In that case the coincidence rate is no longer zero, and the HOM dip is only partial. The present paper does not require a detailed model of every partial-distinguishability profile. It is enough to state the structural principle: complete bunching occurs only when the two-photon state satisfies the framework’s indistinguishability criterion, and departures from that criterion suppress the cancellation.

The HOM benchmark shows that the framework handles a second nontrivial quantum-optical regime beyond Bell. The same junction rule used for single-photon passive networks also governs the two-photon beamsplitter benchmark; exchange symmetry from Paper III has direct observable consequences in the radiation program; and the coherence story is sharpened, since indistinguishability at the two-photon level is the same structural kind of condition as branch coherence at the one-photon level.

HOM bunching therefore follows in the present framework without new primitives: bosonic exchange symmetry, the junction rule, and Born readout together force cancellation in the coincidence sector for indistinguishable photons at a 50/50 splitter.

With that result in place, the next section turns from transport and correlation to angular and topological structure.

9 OAM benchmark

The preceding benchmarks treated polarization, branching, correlation, and indistinguishable two-photon interference. The next question is whether the same framework also carries a nontrivial angular and topological structure on the radiation side. The orbital-angular-momentum benchmark addresses that question.

The present section establishes a layered result. The compact fiber forces integer quantization of angular modes through coherent closure on the compact transverse cycle. At the directly fiber-topological level, it resolves coarse residue structure: parity and angular class. The specific

integer value of ℓ is then resolved in the Paper III representational layer through continuous phase winding. The section does not claim a purely fiber-level distinction of every integer ℓ without the representational layer.

9.1 Primitive angular structure

The starting point is the compact transverse organization of the radiation mode. The primary magnetic cycle is modeled by the finite cycle $\mathbb{Z}_4$, with generator g acting as a quarter-turn on the transverse structure. The covering structure developed in the earlier OAM work supplies the coarser $\mathbb{Z}_2$ holonomy result, distinguishing trivial from nontrivial parity classes under loop transport around the beam axis.

The present question is finer: why should orbital angular content be quantized in integer units at all?

The answer is that a radiation mode is a coherent-return structure. It is not merely any formal phase assignment on the transverse cycle. To count as an admissible mode, its angular response must be compatible with the closure of the cycle it inhabits.

9.2 Coherent closure on the compact transverse cycle

Let χ_ℓ be the angular character carried by a mode under the quarter-turn generator g . A definite angular mode transforms as

$$\chi_\ell(g) = e^{i\ell\pi/2}.$$

One full transverse traversal is four quarter-turns, so

$$g^4 = e,$$

the identity element of the cycle.

The framework-native closure condition is

$$\chi_\ell(g^4) = 1.$$

This is not the standard wavefunction single-valuedness condition on a continuous circle. It is the requirement that a radiation mode be a coherent-return structure compatible with the compact transverse cycle on which it lives.

Substituting the character gives

$$\chi_\ell(g^4) = \chi_\ell(g)^4 = (e^{i\ell\pi/2})^4 = e^{i2\pi\ell} = 1.$$

Hence

$$\ell \in \mathbb{Z}.$$

A concrete example makes the closure argument more transparent. For $\ell = 2$,

$$\chi_2(g) = e^{i2\pi/2} = e^{i\pi} = -1.$$

A full transverse traversal applies four quarter-turns, so

$$\chi_2(g^4) = (-1)^4 = 1,$$

and the mode closes compatibly with the cycle. In this case the angular residue class is 2 mod 4, while the parity class is even. The same logic applies to any integer ℓ : closure fixes integrality, while the finite cycle records only the corresponding residue structure.

9.3 What the fiber resolves directly

The closure argument forces $\ell \in \mathbb{Z}$, but the directly fiber-topological structures are finite. The compact fiber therefore does not, by itself, distinguish every integer ℓ separately. Three levels must be separated.

Parity level

At the parity level, the fiber resolves $\ell \bmod 2$ through the covering holonomy. The covering holonomy yields the coarsest topological shadow,

$$W \in \{+1, -1\}, \quad W = (-1)^\ell.$$

Angular class level

At the angular-class level, the compact transverse cycle resolves $\ell \bmod 4$. The primary magnetic cycle $\mathbb{Z}_4$ distinguishes angular classes through its characters,

$$\chi_\ell(g) = e^{i\ell\pi/2}.$$

Because

$$\chi_{\ell+4} = \chi_\ell,$$

the compact transverse cycle directly resolves only the residue class modulo four.

Full integer level

At the full integer level, the closure condition forces $\ell \in \mathbb{Z}$. But a finite cycle does not distinguish ℓ from $\ell + 4$, $\ell + 8$, and so on. That finer distinction is completed in the representational layer's continuous phase-winding structure. The division of labor should therefore be stated plainly: the fiber forces integer quantization, the fiber directly yields the coarse residue structure, and the representational layer resolves which specific integer in that class is realized.

9.4 Representational completion

In the Paper III layer, branch transport and angular phase are represented continuously. Once that phase structure is available, different winding rates distinguish $\ell = 1$ from $\ell = 5$, $\ell = 9$, and so on, even though all of these modes share the same $\mathbb{Z}_4$ residue class at the directly fiber-topological level. The correct claim is therefore layered. Integrality is forced by coherent closure on the compact transverse cycle. Parity and angular class are directly fiber-topological. The specific integer mode number is then resolved by representational completion. This is stronger than the earlier parity-only result, but more disciplined than claiming that the fiber alone distinguishes every integer ℓ .

9.5 Split-and-recombine consequence

If one mode is split into two paths around opposite sides of the beam axis and recombined at a common endpoint, then the sign ratio of the two contributions equals the holonomy:

$$\frac{s_Q^\alpha}{s_Q^\beta} = W(\gamma).$$

This yields the coarsest operational consequence of the OAM structure. Trivial holonomy gives sign-consistent recombination, whereas nontrivial holonomy gives sign-opposed recombination. The split-and-recombine result is therefore the parity-level readout shadow of the richer angular organization established above.

9.6 Selection rule

At the benchmark level used in this paper, the principal explicit selection consequence remains the parity rule

$$W = (-1)^{\Delta\ell}.$$

A clean parity-violating OAM transfer benchmark would therefore count directly against the framework’s angular and topological structure.

The OAM benchmark does more than reproduce a familiar angular mode label. Standard treatments usually begin with the continuous angular-mode family itself. The present framework instead derives integer quantization from coherent closure, derives parity and angular class from fiber topology, and completes the full integer resolution in the representational layer. The compact-fiber framework therefore supports integer-quantized angular modes through coherent closure on the compact transverse cycle, with parity and angular residue class directly fiber-topological and the specific integer ℓ resolved through representational completion.

The next section turns from angular structure to the radiation–matter interface, where the framework must account for thresholded transfer rather than passive propagation alone.

10 Photoelectric benchmark

The preceding sections treated the passive radiation benchmark regime: polarization, interference, correlation, two-photon bunching, and angular/topological structure. The photoelectric benchmark turns to radiation–matter transfer. It tests whether the same framework can account for a thresholded absorption event rather than only propagation, recombination, and readout.

The standard benchmark to be recovered is familiar. Emission occurs only above a threshold frequency; once threshold is met, emission is effectively immediate; the maximum electron energy is set by frequency rather than intensity; and intensity changes the emission rate rather than the per-electron excess energy. The claim of the present section is not that every microscopic detail of photoemission has already been derived from first principles. For the atomic comparison class, gas-phase photoelectron–photoion coincidence spectroscopy provides the relevant experimental benchmark [1]. The claim is narrower: the event architecture that produces these four features follows from one per-mode transfer scheme, and that architecture is especially strong for atomic targets.

It is useful at the outset to distinguish the structural, representational, and mixed parts of the benchmark. That distinction is summarized in [table 2](#).

The section is therefore not simply a restatement of Einstein’s formula. The structural content lies on the radiation side and the target side before conventional units are applied. The representational layer then expresses the same event in standard energy–frequency language, while the threshold bridge $\nu_{\Phi} = IE/h$ connects the target-side binding scale to the radiation-side mode content.

The decisive structural step is the event model itself. The benchmark treats photoelectric emission as a single radiation-mode-to-electron transfer event at the radiation–matter interface, in which the full Level 1 vibrational content of one minimal coherent-return unit is delivered to one bound electron. Within the present benchmark model, this excludes a gradual pooling picture in

Type	Content
Structural	one mode $\rightarrow$ one electron event; threshold admissibility; excess as incident content minus binding cost; intensity as mode flux; immediacy via per-mode event check
Representational	$E = h\nu$; $K_{\max} = h\nu - IE$; rate and cross-section language
Mixed	$\nu_{\Phi} = IE/h$

Table 2: Structural, representational, and mixed components of the photoelectric benchmark.

which many sub-mode fragments accumulate at one electron until the threshold is reached. The section does not claim a general theorem about all possible microscopic transfer descriptions. It claims that, in the benchmark regime treated here, thresholding and immediacy are both explained by the same per-mode event structure.

The target contributes a binding closure scale. If the weakest bound electron relevant to the benchmark has threshold ν_{Φ} , then the structural threshold condition is simply

$$\nu \geq \nu_{\Phi}.$$

An incident radiation mode carries recurrence content set by its Level 1 scale ν , and emission occurs only when that content is admissible for the target’s escape channel. This is the photoelectric analogue of the admissibility logic that has already appeared in earlier sections: the framework determines the event through structural compatibility rather than through a benchmark-specific extra law.

At this point the distinction between atomic and solid-state targets becomes essential. For atomic photoionization, the threshold is internally supplied by Paper II. If IE denotes the atomic ionization energy derived there, then

$$\nu_{\Phi} = \frac{IE}{h}.$$

This is the strongest version of the benchmark because both sides of the threshold relation are framework-internal. The radiation branch supplies the incident mode content; the atomic branch supplies the binding scale. For solid-state photoelectric emission, by contrast, the framework explains the event architecture and the threshold behavior but does not yet derive the work function Φ from first principles. The section therefore treats atomic thresholds as the strongest cross-module case and solid-state thresholds as architecturally explained but still quantitatively empirical.

Once the threshold is fixed, the frequency dependence follows immediately. A single mode carries one Level 1 recurrence scale ν , and the event couples that mode to one electron. After the binding cost has been paid, the remaining content is the excess. Structurally, that excess is $\nu - \nu_{\Phi}$; representationally, it is written as

$$K_{\max} = h(\nu - \nu_{\Phi}).$$

For atomic targets, where $\nu_{\Phi} = IE/h$, this becomes

$$K_{\max} = h\nu - IE.$$

The standard frequency–energy law is therefore not merely reproduced numerically. It is tied to the one-mode event structure itself: each event carries the content of one radiation mode, and the excess is what remains after subtraction of the target-side binding cost.

The intensity dependence is different for the same reason. At fixed frequency, increasing the intensity increases the number of incident modes per unit time and area, but it does not change the

content carried by each mode. Frequency therefore controls per-event content, whereas intensity controls the rate at which admissible events are available. This is the framework’s structural account of the standard frequency/intensity split in the photoelectric effect.

The same event picture also explains the effective immediacy of emission. In an accumulation picture, one would expect delay because the target would need time to gather enough energy from the incident wave. In the present model, the decision is made per mode at the interface. Each incident mode either satisfies the threshold admissibility condition or it does not. Thresholding and immediacy therefore arise from the same structural feature rather than from two separate empirical rules.

A brief pair of atomic examples is enough to show how the cross-module bridge operates. Hydrogen is the cleanest case because the threshold is exact in the atomic language of the framework. With $IE = 13.598$ eV,

$$\nu_{\Phi} = \frac{IE}{h}, \quad \lambda_{\Phi} \approx 91.2 \text{ nm}.$$

Thus incident light at 100 nm lies below threshold, since $100 \text{ nm} > 91.2 \text{ nm}$ and therefore corresponds to a frequency below ν_{Φ} , and does not produce emission, whereas light at 80 nm lies above threshold and yields positive excess. Sodium is useful as a benchmark-format validation case. With threshold corresponding to $IE \approx 5.139$ eV,

$$\lambda_{\Phi} \approx 241.3 \text{ nm},$$

so 250 nm lies below threshold, 230 nm lies slightly above it, and 200 nm yields larger excess. The scientific point of these examples is not the arithmetic itself, but the framework-internal chain

$$\text{Paper II ionization energy} \longrightarrow \nu_{\Phi} = \frac{IE}{h} \longrightarrow \text{photoelectric threshold and excess law}.$$

These examples belong to the same experimental comparison class probed by gas-phase photoelectron-photoion coincidence spectroscopy [1].

The photoelectric benchmark therefore extends the paper beyond passive optical structure. The compact-fiber radiation architecture models photoelectric emission as a per-mode radiation-matter transfer event, from which thresholding, immediacy, the frequency-energy law, and the intensity-rate separation follow; for atomic targets, the threshold is internally fixed by the Paper II ionization energy, while solid-state work functions remain empirical inputs.

With the benchmark suite now in place, the next section turns to the scaling layer used across that suite: the Planck relation as the bridge between recurrence-based natural units and conventional energy-frequency units.

11 Planck scaling and unit bridge

The benchmark sections use the standard energy-frequency bridge

$$E = h\nu$$

and, in the atomic photoelectric case, the threshold relation

$$\nu_{\Phi} = \frac{IE}{h}.$$

Within the present paper, these relations function as the natural-to-conventional unit bridge between recurrence-based radiation content and standard measurement units.

The broader compact-fiber program motivates a structured decomposition of this unit bridge. In that setting, the proportionality in $E = h\nu$ is not treated as a primitive radiation law inserted independently for each benchmark, but as the conventional-unit expression of a more basic recurrence-content relation. The full derivation, input ledger, and numerical evaluation of that decomposition are recorded in Appendix A.

For the benchmark regime established in the main text, the point is narrower. The suite does not require a separate scaling constant for each phenomenon. The same bridge appears in general energy assignments, in the atomic photoelectric threshold relation, and more broadly whenever recurrence-based natural content is expressed in conventional energy–frequency units.

The next section provides a compact translation between the paper’s radiation-side language and the standard quantum-optical objects used across the benchmark regime.

12 Mapping section

The compact-fiber radiation framework is not intended to replace all standard quantum-optical language with unfamiliar terminology. Its purpose is narrower and more useful: to show that the benchmark objects used in standard practice can be grounded in a more primitive radiation architecture and then recovered in a shared representational/readout layer. A compact mapping between the two idioms is therefore helpful, but that mapping should not be mistaken for the substance of the paper. Its role is to keep the relation between the standard objects and the compact-fiber objects explicit while the later sections develop the comparative and predictive claims.

The correspondence used throughout the paper is summarized in table 3. A polarization state corresponds to a Level 2/3 radiation mode; an analyzer basis corresponds to a readout context; a path phase corresponds to a branch transport factor; a beamsplitter matrix corresponds to the junction rule; a Born probability corresponds to the shared readout law; the Bell singlet corresponds to the unique opposed pair state; OAM parity and angular class correspond to covering holonomy and the compact transverse cycle; the photoelectric threshold corresponds to a binding closure scale; and the Planck relation corresponds to the unit bridge between recurrence content and conventional energy units.

What the table fixes is the level at which the comparison with legacy theory is being made. Standard treatments usually begin directly with the objects listed in the left-hand column. The present paper claims that, within the benchmark regime treated here, those objects are not independent primitives. They are representational forms of a smaller set of radiation-side structures. This is why the table belongs in the main text rather than in an appendix. Without it, the reader is forced to translate repeatedly between the standard benchmark language and the compact-fiber language while the argument is already moving.

At the same time, the table should not be read as a claim that the standard formalism can simply be discarded. The representational layer remains necessary. What changes is its status. It is no longer treated as the sole starting point. The present framework claims that the representational objects can be physically grounded rather than merely postulated. That claim is stronger in some sectors than in others. In polarization, interference, Bell, and HOM, the benchmark law is recovered fully within the stated regime. In the photoelectric case, the event architecture is recovered strongly, but some target-side quantitative inputs remain empirical. In the OAM case, the compact fiber directly supplies coarse topological and angular residue structure, while the representational layer resolves the specific integer ℓ .

The point of the mapping section is therefore practical and conceptual at once. Practically, it fixes the translation between the paper’s radiation-side language and the standard benchmark

Standard object	Compact-fiber / radiation object	Role in the paper
polarization state	Level 2/3 radiation mode	single-arm readout
analyzer basis	readout context	projection / basis choice
path phase	branch transport factor	interference and network propagation
beamsplitter matrix	junction rule	passive linear optical network structure
Born probability	shared readout rule	all benchmark readout laws
coherence / decoherence criterion	readout-class matching under detector projection	interference, HOM, and falsifiability conditions
Bell singlet	unique opposed pair state	correlation benchmark
OAM parity / angular class	covering holonomy / compact transverse cycle	topological and angular benchmark
photoelectric threshold	binding closure scale	radiation–matter transfer benchmark
Planck relation	unit bridge between recurrence content and conventional energy units	scaling layer across the paper

Table 3: Translation between standard quantum-optical objects and the compact-fiber radiation objects used in the present paper.

objects. Conceptually, it makes clear that the comparison with legacy theory is not being carried out at the wrong level. The framework does not deny the usefulness of the standard objects; it denies that they must remain disconnected primitives. The next section takes that claim one step further by stating explicitly what the framework adds beyond this translation alone.

13 What the framework adds

A paper of this kind must be judged against current legacy theory and research practice, not against an empty background. The relevant comparison class is the usual benchmark-by-benchmark treatment of the same radiation phenomena. Against that comparison class, the present paper makes a specific claim: the compact-fiber radiation framework does more than reproduce the benchmark laws; it does so with fewer independent assumptions and with greater structural continuity than the usual compartmentalized treatments of the same phenomena.

The earlier sections already established that the framework recovers the benchmark suite in the stated regime. The point of the present section is to state clearly what is gained by doing so in this way. The gain is not identical in every benchmark. In some cases it is a matter of grounding structures that standard treatments typically leave primitive. In others it is the removal of a phenomenon-specific extra law. In still others it is a structural selection rule or a more economical scaling account. The compact summary is given in [table 4](#).

The status column is not decorative. It is included to prevent the comparison table from

flattening distinctions that the paper has already made explicit in [section 2](#). In particular, the OAM and photoelectric rows should be read with the same caution as in the benchmark sections themselves, and the scaling row is included as a structured bridge rather than as a main-text theorem-grade derivation. The point is not to discard standard representational practice. The framework does not try to replace projection, Hilbert-space structure, or Born probabilities with an unrelated computational language. Its claim is different. It claims that the standard benchmark objects can be grounded in a smaller set of radiation-side structural ingredients. In that sense, the present work is not valuable because it avoids standard mathematics. It is valuable because it explains why the same mathematical layer reappears across phenomena that legacy practice often presents separately.

This is clearest where benchmark-specific extra laws disappear. In the polarization benchmark, the analyzer law is not inserted independently. In the interference benchmark, no separate wave law is added once coherent branch recombination and Born readout are in place. In the Bell section, the cosine law is not introduced as a special correlation rule; it follows from the unique opposed pair state together with the shared readout layer. In the HOM section, no separate bunching rule is introduced beyond exchange symmetry, junction structure, and the same readout architecture. In the photoelectric section, thresholding, immediacy, and the frequency/intensity distinction no longer appear as disconnected empirical rules; they arise from one event architecture.

The framework also adds structural content that standard benchmark descriptions often leave ungrounded. The OAM benchmark is the clearest example. Standard treatments typically begin with the angular mode family itself. The present framework instead derives integer quantization through coherent closure on the compact transverse cycle and identifies the coarse residue structure directly at the fiber-topological level. The coherence criterion is another example. Standard decoherence language often treats the system/environment split as externally chosen; here the distinction is tied to readout-class matching. The scaling discussion adds the corresponding bridge result: $\hbar$ is not primitive but appears as the unit bridge between recurrence-based natural content and conventional units, with its fuller structural decomposition recorded in [Appendix A](#).

The comparison with legacy theory is therefore favorable in a specific and limited sense. Within the benchmark regime treated here, the compact-fiber radiation framework is more parsimonious because it requires fewer independent starting assumptions, more structurally continuous because one architecture and one shared readout layer span the suite, and more derivationally grounded because several benchmark ingredients that standard practice leaves primitive are here derived or physically anchored. That is the paper’s main theoretical gain. The benchmark laws are not only reproduced; they are reproduced from less, and in a more unified way, than in the usual benchmark-by-benchmark presentation.

The next section turns from that comparative claim to explicit commitments. If the framework is to count as more than a structurally economical reformulation, it must state what would falsify it and what novel constraints follow from it within the benchmark regime.

14 Predictions and falsifiability

A theory paper is not strengthened simply by recovering familiar benchmark results more economically. It must also state what would count against it and what structural or quantitative consequences it commits itself to. The present section makes those commitments explicit for the compact-fiber radiation benchmark regime.

The predictive content of the framework has two distinct forms. In one form, the framework commits itself to the standard benchmark laws within the regime stated earlier in the paper. These

are not the strongest novelty claims, but they are still binding: if the benchmark regime fails to behave in the way the framework says it must, the framework fails in that sector. In the second form, the framework makes stronger structural commitments that are not exhausted by the statement that a known benchmark law has been reproduced. Those include selection rules, quantization claims, closure conditions, and the structure of the scaling bridge itself.

The first class of commitments can be stated compactly. Within the stated regime, the framework must recover the Bell polarization law

$$E(\alpha, \beta) = -\cos 2(\alpha - \beta),$$

the CHSH-optimal value

$$|S| = 2\sqrt{2},$$

the HOM coincidence suppression

$$P_{\text{coinc}} = 0$$

for indistinguishable symmetric input at a balanced junction, the interference law

$$P = |c_\alpha + c_\beta|^2$$

under coherent branch recombination, and the fourfold photoelectric pattern consisting of thresholding, immediacy, frequency control of per-electron excess, and intensity control of rate. These are not claimed as new anomalies. They are claimed as benchmark-regime commitments. If a clean benchmark inside the paper's stated domain fails to show them, then the framework is not merely less economical than hoped; it is wrong in that sector.

The stronger predictive content lies in the structural constraints that the framework imposes. The most explicit of these is the OAM parity selection rule

$$W = (-1)^{\Delta\ell},$$

which turns the OAM result into a falsifiable transfer constraint. In operational terms, a counterexample would be a passive-regime OAM-transfer benchmark in which the incident mode's parity class W is independently identified, the target-side angular change $\Delta\ell$ is independently known, and the observed transfer violates

$$W = (-1)^{\Delta\ell}.$$

Such a result would count directly against the framework's angular and topological structure. Closely related is the claim that coherent closure on the compact transverse cycle forces

$$\ell \in \mathbb{Z}.$$

Within the regime treated here, a benchmark requiring a fractional angular mode number would therefore count directly against the present angular-closure argument. The framework also commits itself to the photoelectric threshold bridge

$$\nu_\Phi = \frac{IE}{h}$$

when IE is supplied by the Paper II atomic sector. Systematic failure of the atomic threshold mapping would therefore count against one of the paper's strongest cross-module claims.

The same is true of the readout-class coherence criterion. The paper does not treat coherence and decoherence as determined by an arbitrary system/environment split. It treats them as determined

by whether the branches remain matched in readout class except for phase-carrying structure. A clear mismatch between that criterion and observed coherence behavior in the benchmark regime would therefore count against the framework. Finally, the Planck section commits the framework to the structural form

$$h = \frac{m_n c S_n^2}{R} (1 + s)^{1/3}$$

once the two empirical anchors are fixed. Failure of that form under a correct calibration chain would count against the scaling layer itself.

The point of stating these commitments is not to overstate the paper’s novelty. It is to make clear that the framework is exposed to failure where it claims success. The paper is not falsified because some phenomenon outside the stated regime requires additional machinery. It is falsified when a phenomenon inside the stated regime fails to behave in the way the framework says it must. This is why the regime statement in [section 2](#) is part of the paper’s logic rather than a protective aside.

The broader research program contains other predictive threads, including attosecond timing consequences, gravity-side distinguishing results, and additional atomic or ionic spectroscopic predictions. Those are not developed here because they would dilute the present paper’s focus. Their relevance is only to show that the radiation benchmark regime treated here is not an isolated reconstruction, but part of a larger program that also makes externally testable commitments.

The compact-fiber radiation framework is therefore not only more economical in the benchmark regime. It is also more exposed there. That exposure is part of the paper’s claim rather than a liability external to it. The next section states the boundaries of that claim explicitly, so that the framework is judged neither more broadly nor more narrowly than the present paper warrants.

15 Boundaries, exclusions, and follow-on problems

A paper of this breadth can fail in two opposite ways. It can become so narrow that the reader cannot tell what the framework is meant to cover, or so diffuse that it appears to claim every adjacent problem without carrying the necessary derivation. The present paper avoids the second error by stating its boundaries explicitly.

The most important boundary is that the paper is about the compact-fiber radiation benchmark regime. It does not attempt a complete radiation theory, nor does it claim a full treatment of all optical or many-photon phenomena. Within that regime, the benchmark suite is broad: polarization, interference, Bell correlation, HOM bunching, angular structure, photoelectric transfer, and the scaling bridge to conventional units. Outside that regime, the paper is intentionally selective.

Several topics are therefore excluded from the present treatment. The paper does not attempt a purely fiber-level distinction of every integer ℓ without the representational layer. It does not attempt a first-principles derivation of solid-state work functions. It does not treat nonlinear optics, gain media, lasing, full Fock-space or variable-photon-number structure, antibunching, delayed-choice quantum erasure, or full spin-orbit coupling. It also does not claim internalization of the mass anchor in the Planck scaling layer. Some of these problems are outside the present benchmark regime. Others are natural continuations of the framework, but would weaken the present paper if developed here only partially.

These exclusions should not be read as missing prerequisites for the paper’s central claim. They are follow-on problems. The main benchmark suite does not depend on resolving them first. What remains open is narrower. One open question is whether the fiber alone can distinguish all integer ℓ sectors without representational completion. The present paper shows that coherent closure on the

compact transverse cycle forces integer quantization, and that the fiber directly yields coarse class structure, while the specific integer mode number is resolved in the representational layer. Whether the fiber can be enriched so as to distinguish every integer directly is a genuine continuation problem rather than a precondition for the present result.

A second open problem concerns solid-state thresholds. The photoelectric benchmark is strongest for atomic targets, because the threshold can be tied directly to the Paper II ionization energy. Extending that level of internal derivation to work-function physics remains a separate task. A third concerns broader source-side structure. The present paper treats passive linear radiation and the core radiation–matter transfer benchmark. Problems such as antibunching, gain, and more elaborate source statistics lie beyond that scope. A fourth concerns the mass anchor in the scaling layer. The Planck section reduces the scaling problem to two dimensionally independent empirical anchors. Whether the mass side can eventually be internalized elsewhere in the broader framework remains open.

These boundaries strengthen the paper rather than weaken it. A foundational reference is more useful when its domain is explicit. The present work therefore makes a strong but bounded claim: within the compact-fiber radiation benchmark regime, one architecture together with one shared representational/readout layer suffices to recover the benchmark suite and its structural consequences. Outside that regime, further work is required and is not preemptively claimed here.

The conclusion can therefore restate the paper’s central challenge claim without widening the scope beyond the benchmark regime actually established.

16 Conclusion

The paper began from a specific claim: that a broad benchmark class of radiation phenomena can be recovered from one compact-fiber radiation architecture together with the already-established Paper III representational/readout layer, without introducing disconnected phenomenon-specific rules. The benchmark suite treated here was chosen to span the main kinds of structure the radiation program must handle in its present regime: single-arm readout, coherent branching and recombination, non-factorable correlation, indistinguishable two-photon interference, angular and topological structure, radiation–matter transfer, and the scaling bridge to conventional units.

That claim has been made concrete across the benchmark suite. Polarization is recovered within the shared representational/readout layer as the image of Level 2/3 transverse mode structure. Interference follows from coherent branch recombination under the radiation network calculus. Bell correlation follows from the unique opposed pair state together with the same shared readout layer. HOM bunching follows from exchange symmetry, the junction rule, and the same general readout structure. The OAM benchmark shows that compact transverse closure forces integer quantization of angular modes while the direct fiber topology supplies the coarse class structure. The photoelectric benchmark shows that the framework also handles thresholded radiation–matter transfer, not only passive optical transport and readout. The scaling relation $E = h\nu$ functions throughout as the bridge between recurrence-based natural content and conventional energy–frequency units, and its broader structural decomposition has been recorded separately in Appendix A.

The paper’s stronger claim is not merely that it reproduces the standard benchmark laws. We claim that it does so with fewer independent assumptions and with greater structural continuity than the usual compartmentalized treatments of the same phenomena. In the benchmark regime treated here, the compact-fiber radiation framework compares favorably with current legacy theory because one radiation architecture and one shared representational/readout layer replace multiple phenomenon-specific starting assumptions.

The paper also makes explicit predictive and falsifiable commitments. Some are benchmark-regime commitments: if the framework is right in the stated regime, it must recover the Bell law, the CHSH optimum, the HOM dip, the interference law, and the photoelectric event pattern. Others are stronger structural claims: the OAM parity selection rule, the coherent-closure origin of integer angular quantization, the readout-class coherence criterion, the atomic photoelectric threshold bridge, and the existence of a structured scaling bridge relating recurrence-based natural content to conventional units. These commitments are what make the paper a theory paper rather than a more economical redescription of familiar results.

The paper’s boundaries are part of its result. It does not claim complete radiation theory, first-principles solid-state work functions, nonlinear optics, full Fock-space treatment, antibunching, delayed-choice phenomena, or a purely fiber-level distinction of every integer angular mode. Those remain outside the present benchmark regime or are deferred as follow-on problems. This scope discipline is what allows the present result to function as a foundational reference for the regime it actually treats.

We therefore conclude with the paper’s central challenge claim: within the radiation benchmark regime treated here, the compact-fiber radiation framework provides a more parsimonious, more structurally unified, and more derivationally grounded account than the usual legacy patchwork treatment. One compact-fiber radiation architecture, together with the already-established Paper III representational/readout layer, accounts for the benchmark suite through a common structural backbone while making explicit structural and quantitative commitments that render the framework falsifiable.

A Planck scaling derivation and input ledger

This appendix records the full scaling decomposition underlying the Planck relation in the compact-fiber radiation program. Its role is supplementary to the benchmark suite established in the main text. The body of the paper uses the unit bridge $E = h\nu$ and, in the atomic photoelectric benchmark, $\nu_\Phi = IE/h$. The present appendix gives the structural derivation, the input ledger, and the numerical interpretation supporting that bridge.

A.1 Structural role of the Planck relation

Within the compact-fiber radiation architecture, $E \propto \nu$ is structurally forced. The Level 1 mode is an indivisible coherent-return unit whose only intrinsic scale is its vibrational frequency ν . Energy is the Paper III representational image of this scale. Linearity follows from additivity: two modes at ν carry twice the content of one. In the framework’s natural units, the relation is therefore $E = \nu$. The content of the claim is not that this is a tautology, but that linearity in ν , rather than some nonlinear function of recurrence scale, is forced by the additivity of Level 1 mode content. The proportionality constant h enters only when recurrence content is expressed in conventional units such as Hertz and ergs.

A.2 Canonical form

The canonical form used in the broader scaling analysis is

$$h = \frac{m_n c S_n^2}{R} (1 + s)^{1/3},$$

where S_n is the natural unit of space, m_n the natural unit of mass, c the speed of light, R the interregional ratio, and $(1 + s)^{1/3}$ the secondary mass correction. The framework fixes this form algebraically; the empirical anchors set the scale.

A.3 Structural ingredients

Two parts of the formula are structural in the sense relevant here.

The first is the interregional ratio

$$R = 128 \times \frac{11}{9} = 156.444 \dots$$

This factor is structural and combinatorial. In the derivation inherited from the underlying RST analysis, the rotational contribution comes from two 2-dimensional magnetic rotations and one 1-dimensional electric rotation, giving

$$4 \times 4 \times 8 = 128.$$

In compact-fiber language, this is read through the three cyclic factors of

$$F \cong Z/4Z \times Z/4Z \times Z/8Z,$$

so that the combinatorial count records the admissible rotational organization carried by the two magnetic $Z/4Z$ sectors and the electric $Z/8Z$ sector at the level relevant to the scaling decomposition. The product $4 \times 4 \times 8$ is therefore not an arbitrary invariant of the fiber, but the specific cardinality product associated with those three cyclic sectors in the inherited rotational count. The vibrational contribution then enters as an additional factor $1 + 2/9$, yielding

$$128 \times \frac{11}{9}.$$

Here the term $2/9$ is inherited from Nehru's vibrational degree-of-freedom count [5]: the two magnetic systems contribute an additional $2/9$ relative to the rotational total once the corresponding vibrational possibilities are included.

The second is the secondary mass correction

$$(1 + s)^{1/3}.$$

Here the exponent $1/3$ is fixed by the dimensional relation between energy and mass in the underlying RST scaling: energy carries dimensions t/s , whereas mass carries dimensions $(t/s)^3$. A mass-side correction therefore enters the energy-side scaling at the cube root. What is not reproduced here in independent detail is the internal composition of the secondary mass quantity s . The present appendix records only the fact that the full decomposition carries the factor $(1 + s)^{1/3}$, with $s = 0.00697$ in the inherited evaluation (magnetic and electric mass components combined).

A.4 Empirical anchors

The framework requires two dimensionally independent empirical anchors to set the natural-unit scale if the Paper III derivation of α is accepted; otherwise the empirical count rises by one. The first anchor is spectroscopic and fixes the length side through S_n . The second is mass-side and fixes the scale of m_n . The speed of light is exact by convention. The status of α should therefore be stated separately: if the Paper III derivation is taken as established, then α is not an additional

empirical input at this stage and the anchor count remains two; if not, the count rises to three. The scaling decomposition itself does not depend on how that bookkeeping choice is made; only the headline parsimony count does.

More concretely, S_n is set from the spectroscopic side through the Rydberg scale, while m_n is set through the mass or gravitational side. This is the minimum expected anchor pair for a natural-unit framework that does not already derive the absolute mass scale internally.

A.5 Derivation trace

The derivation proceeds through the trace

$$h = \frac{E_n T_n S_n}{R} (1 + s)^{1/3},$$

together with

$$T_n = \frac{S_n}{c}, \quad E_n = m_n c^2.$$

Substituting gives

$$E_n T_n S_n = m_n c^2 \left(\frac{S_n}{c} \right) S_n = m_n c S_n^2,$$

hence

$$h = \frac{m_n c S_n^2}{R} (1 + s)^{1/3}.$$

Attribution for the original Planck-constant and interregional-ratio derivations belongs to Nehru [4, 5]; the compact-fiber correspondences, canonical rewriting, and calibration interpretation are part of the present work.

A.6 Numerical status

Using Larson’s historical natural-unit values, the canonical form gives

$$h = 6.6256 \times 10^{-27} \text{ erg} \cdot \text{s}.$$

The modern defined value is

$$h = 6.62607015 \times 10^{-27} \text{ erg} \cdot \text{s}.$$

Within the present framework, this historical mismatch is attributed to the empirical anchors used in the older natural-unit evaluation rather than to the structural factors R and $(1 + s)^{1/3}$. Evaluation with modern-precision anchors is deferred to a separate calibration note. The purpose of the present appendix is therefore to record the structural form, the substitution chain, the historical numerical evaluation, and the location of empirical dependence, rather than to elevate the calibration question into a separate main-text claim.

A.7 Reuse across the radiation program

The scaling bridge is reused across the radiation program rather than introduced separately for different phenomena. In the photoelectric benchmark it appears as

$$\nu_\Phi = \frac{IE}{h},$$

while more generally it appears as

$$E = h\nu.$$

It therefore functions as the common bridge from recurrence-based natural content to conventional energy–frequency units across the benchmark regime.

A.8 Input ledger

The input ledger may be summarized as follows. The Rydberg constant R_∞ provides the spectroscopic anchor setting S_n . The mass or gravitational side provides the anchor setting m_n . The speed of light c is exact. The interregional ratio $R = 128 \times 11/9$ is structural. The factor $(1 + s)^{1/3}$ enters through the full structural decomposition, with its cube-root exponent fixed dimensionally and its internal s -composition inherited from Larson’s mass analysis. The quantities T_n and E_n are then derived from S_n , m_n , and c , and h follows from the canonical form above. In that sense the framework reduces the scaling problem to the expected pair of empirical anchors while fixing the remaining form algebraically.

A.9 Established and open points

For the purposes of the present paper, the established points are these: $E \propto \nu$ is structurally forced by the radiation architecture; h is treated as a natural-to-conventional unit conversion factor rather than as a primitive radiation constant; the canonical form above records the structured decomposition used in the broader program; $R = 128 \times 11/9$ is combinatorial; and the scaling problem reduces to two empirical anchors at the present stage. The main open point is whether the mass anchor can eventually be internalized, for example through a successful derivation of the relevant mass-ratio structure elsewhere in the program.

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Benchmark	Status	Standard treatment typically assumes	Framework derives / grounds
Polarization / Malus	Closed	analyzer law	analyzer law recovered within the shared representational/readout layer, with the polarization state interpreted as the image of Level 2/3 transverse structure
Interference	Closed	amplitude superposition rule applied case-by-case	branch transport, readout-class coherence, and Born readout
Bell	Closed	singlet state, Born rule, and analyzer formalism	pair state forced by opposition, analyzers grounded in Level 2/3 structure, and reused Born layer
HOM	Closed	bosonic symmetry and beamsplitter algebra	the same, together with junction phase J interpreted through the compact-fiber quarter-turn structure and readout-class indistinguishability
OAM	Closed with representational completion	integer ℓ modes and parity structure taken from standard angular formalism	integrality from coherent closure, coarse residue structure from fiber topology, and full integer resolution supplied in the representational layer
Photoelectric	Closed at benchmark level	$E = h\nu$ quanta and Einstein threshold law	per-mode transfer architecture, target-side binding closure, and the cross-module atomic threshold bridge
Planck scaling	Closed at benchmark level	h primitive	structured scaling bridge with two empirical anchors; full decomposition recorded in Appendix A

Table 4: Comparison between standard benchmark starting points and the structures grounded by the compact-fiber radiation framework, with statuses matching the claim-strength discipline of [section 2](#).